



Understanding Lock Out / Tag Out

CNPC Chuanqing Drilling Engineering Company Limited (CCDC)





Defining Lockout & Tagout

LOCK OUT



The process of cutting all sources of energy and installing a personal lock at the source.

The physical lock prevents the starting of machinery while it is being cleaned, maintained, adjusted, or repaired.

TAG OUT



The placement of a tag on an energy-isolating device. It serves as a warning of the danger of operating the machine.

**CRITICAL WARNING: A tag indicates 'DO NOT START'.
However, a tag Never replaces a lock**



The Objective : Zero Energy State



Definition: The point at which equipment has been safely isolated from ANY chances of re-energization.

Scope: This includes the release of residual(stored) energy, not just disconnecting power.

LOTO is the tool ; **Zero Energy** is the goal



Know Your Enemy: Hazardous Energy Types

Electrical



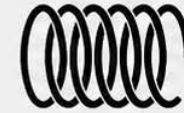
Transmission lines, transformers, circuit breakers, motors (AC/DC).

Kinetic (Motion)



Moving conveyors, flywheels, blades. Energy caused by motion.

Potential (Stored)



Springs, batteries, elevated weights. Stored energy waiting to release.



Hydraulic & Pneumatic

Fluid or compressed air/gas in cylinders and pipes.



Thermal

Heat generated from machines ($>60^{\circ}\text{C}$) or freezing systems ($<5^{\circ}\text{C}$).



Other Hazards

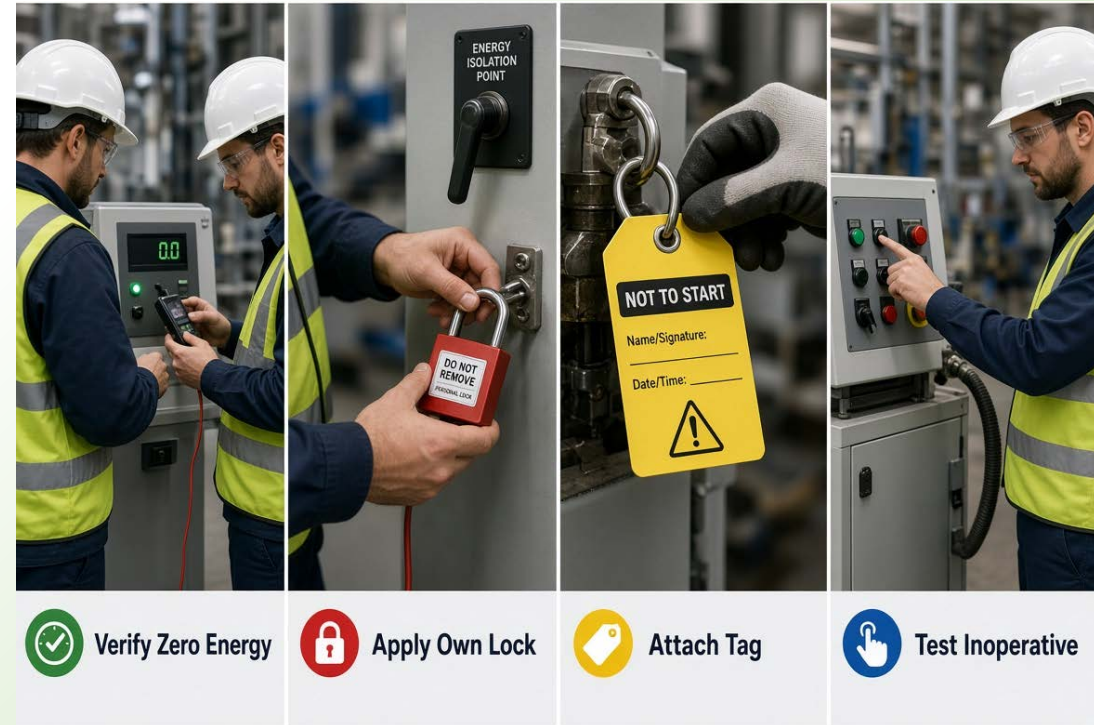
Chemical reactions, Radiation, Magnetic fields, Steam pressure.



Roles & Responsibilities

Employee (LOTO user) Responsibilities

- Verify a competent person to achieved Zero Energy State.
- Lock out the machine using the assigned safety lock(never borrow).
- Apply a tag with: 'Not To Start',Name/Signature, and Date/Time..
- Test that the machine is inoperative before working.





The 6 Steps of LOTO



1. PREPARE



2. SHUTDOWN



3. ISOLATE



4. LOCK & TAG



5. CONTROL



6. VERIFY



Step-1: Preparation & Planning

Understanding the problem

- What needs fixing?
- What hazards are involved?
- Is the equipment-specific LOTO procedure available?

Plan the Work

- Are tools & personnel available?
- Have affected employees been notified

Prepare the Area

- Is the area clear
- Have all disconnect points been located





Step-2 & 3: Stop & Disconnect

Step 2: SHUTDOWN



Use established normal stopping Procedures (e.g. Stop Button, Power Switch)



Involve the worker who normally operates the equipment



Rule: If at a customer site, the customer must shut down the equipment

Step 3: ISOLATE



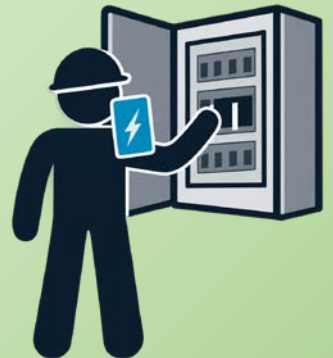
Disconnect from energy sources (close valves, open breakers).



Wear appropriate PPE

ARC FLASH SAFETY

When opening an electrical disconnect, stand to the **RIGHT HAND SIDE** of the panel. Use your **LEFT HAND** to pull the breaker. This protects your body if an arc occurs.





Step-4: Application of Locks & Tags

Rules:

- Use only approved locks & Tags
- Never lend or borrow keys
- Tags must include :
 - I. Warning (“Do not start”)
 - II. Name
 - III. Signature





Step-5: Control Stored Energy

Isolation is not enough. You must neutralize residual energy

Mechanical Motion : Stop motion completely

Gravity : Use blocks & pins for elevated parts

Thermal: Allow equipment to cool down or heat up to safe levels

SPRINGS: Block, tag,& lock compressed springs.

Electrical: Drain power from capacitors.





Step-5: Verify Zero Energy

The Most Important Step: Trust but Verify.



1 Notify affected persons.

2 Attempt to turn equipment ON (Start Button).



3 Test for voltage and check gauges.

**DO NOT START
DO NOT OPEN
DO NOT CLOSE
DO NOT ENERGIZE
DO NOT OPERATE**

4 **CRITICAL:** Return controls to 'OFF' after testing.





Re-Energizing: The Release Procedure

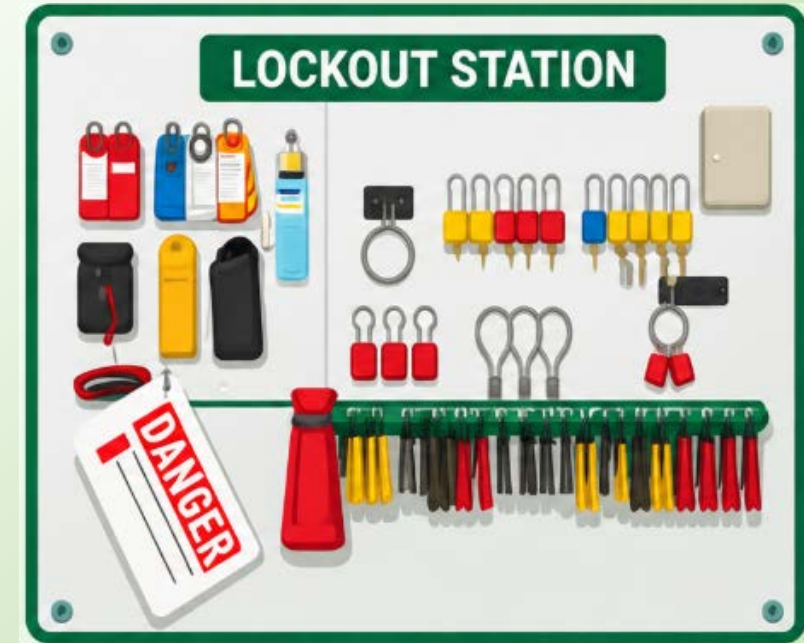
INSPECT: Verify the is safe. Remove tools. Ensure guards are back in place

NOTIFY: Tell all employees the system is going live

UNLOCK: Remove your own lock & tag

RESTORE: Return system functions.

TEST: Re-energize & test equipment





Avoid These Fatal Excuses

It will only take a minute....

Its lunch time, no one is around

The machine is stopped, it does not need a lock



Always Remember

ONE KEY, ONE LOCK, ONE LIFE

Never lend your key, never remove another worker's lock, and never assume a system is safe without testing it yourself.

Zero Energy, Zero Accident